Jesse Coggins

Data Analyst · Data Scientist · Python · SQL · Tableau · Machine Learning
M.S. Data Analytics – Data Science
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SUMMARY

Data analyst and data science candidate with nearly a decade of multi-site operations leadership translating KPI, revenue, labor, and throughput data into reporting, decisions, and process improvement. Public portfolio projects span SQL, Python, statistical analysis, Tableau dashboard development, forecasting, predictive modeling, deep learning, and MLOps workflows built with MLflow, DVC, FastAPI, and Docker. Seeking a remote Data Analyst or Data Scientist role where business context, analytical rigor, and clear communication drive measurable impact.

PROJECTS

Loan Default Classifier *Python · Scikit-learn · Random Forest · Cross-Validation · ROC-AUC*

- Engineered a binary classification pipeline on 148,670 loan records — data wrangling, feature engineering, class imbalance handling via balanced class weighting, and stratified k-fold cross-validation — achieving 84.7% accuracy and 73.7% ROC-AUC
- Analyzed precision-recall tradeoffs and confusion matrix results, framing outputs as a business risk decision framework in a formal analytical report

Customer Churn Analytics Dashboard *Tableau · Exploratory Data Analysis*

- Designed an interactive Tableau dashboard surfacing customer churn patterns and retention risk signals from exploratory data analysis of a telecom dataset
- Published the dashboard as a Tableau Public portfolio artifact, translating churn patterns across contract type, income, geography, and payment method into stakeholder-facing retention insights

Hospital Readmission Analysis *Python · Scikit-learn · Statsmodels · K-Means · PCA*

- Applied three analytical methods to hospital readmission data: Random Forest classification (97.9% accuracy), K-Means clustering for unsupervised patient risk segmentation, and PCA for dimensionality reduction and cluster visualization — demonstrating breadth across supervised, unsupervised, and dimensionality reduction techniques

Flight Delay Prediction Pipeline *Python · FastAPI · Docker · MLflow · DVC · GitLab CI/CD*

- Architected an end-to-end ML engineering pipeline — data versioning (DVC), experiment tracking (MLflow), containerized model deployment (FastAPI + Docker), and automated build, test, and deployment via GitLab CI/CD — replicating production MLOps workflows from data ingestion to live prediction endpoint
- Managed full model lifecycle: reproducible training workflows, experiment tracking and artifact logging in MLflow, and a live REST API serving real-time predictions from a Dockerized FastAPI endpoint

Statistical Analysis & Market Basket Analysis *Python · SciPy · Statsmodels · Mlxtend · Apriori*

- Conducted hypothesis testing on health insurance data — Pearson correlation, one-way ANOVA, Mann-Whitney U, and Chi-Square/Cramér's V — to identify statistically significant drivers of insurance charges
- Applied Market Basket Analysis using the Apriori algorithm on 8,234 retail transactions, surfacing high-lift product association rules (lift = 88.2) to inform cross-promotion strategy

Revenue Time-Series Forecasting *Python · Statsmodels · ARIMA*

- Built and evaluated an ARIMA forecasting model on 731 observations of daily revenue data, applying stationarity testing, parameter selection, and forecast evaluation (RMSE 0.4887; MAE 0.3766)

Relational & Non-Relational Database Design *PostgreSQL · MongoDB · SQL · ETL*

- Designed and implemented a normalized 3NF relational database in PostgreSQL across 5 tables and 100,000 records — including staging table ETL, query optimization via indexes and CTEs, and reusable Views for analytical reporting
- Built a parallel MongoDB implementation with aggregation pipelines and compound index optimization, demonstrating proficiency across both relational and document-oriented database paradigms

Deep Learning: Computer Vision & NLP *Python · TensorFlow · Keras · CNN · Bi-LSTM*

- Trained a Convolutional Neural Network (CNN) for multi-class computer vision classification of plant seedling species (66.6% accuracy) and a Bidirectional LSTM for NLP sentiment analysis (76% accuracy) — applying deep learning across both image and text domains
- Analyzed training dynamics via loss curves and validation accuracy; evaluated generalization to held-out test sets and documented architectural decisions

Operations Research & Mathematical Optimization *Python · PuLP · Linear Programming · MILP*

- Formulated and solved LP and MILP optimization problems, including a multi-node Amazon Air transshipment model with an optimal cost of $200,863.75 using PuLP — demonstrating quantitative modeling skills that extend beyond standard ML into operations research

SKILLS

Languages: Python, SQL
Analytics & BI: Tableau, Pandas, NumPy, Matplotlib, Seaborn · Data Wrangling, Data Cleaning, EDA, KPI Reporting, Dashboard Development, Business Intelligence
Statistical & Machine Learning: Scikit-learn, Statsmodels · Random Forest, Logistic Regression, K-Means, PCA, ARIMA · Hypothesis Testing, Predictive Modeling, Classification, Clustering, Time-Series Forecasting, Association Rule Mining
Deep Learning & NLP: TensorFlow, Keras · CNN, Bi-LSTM · Computer Vision, Sentiment Analysis, Model Evaluation
Data Engineering & Deployment: PostgreSQL, MongoDB, ETL, Docker, FastAPI, MLflow, DVC, GitLab CI/CD · Data Versioning, Experiment Tracking, CI/CD Pipelines

EDUCATION

M.S. Data Analytics – Data Science — Mar 2026
Western Governors University
Coursework: *Machine Learning, Statistical Analysis, Data Mining, MLOps, Deep Learning, Optimization, Database Management*

B.S. Management — May 2024
Jacksonville State University
Summa Cum Laude · Special Honors in Management

A.S. General Studies – Business Administration — Aug 2022
Gadsden State Community College

EXPERIENCE

General Manager — Jan 2022 – Feb 2026
Xtreme Car Wash · Anniston, AL

- Tracked and analyzed daily KPIs — revenue, labor cost %, and customer throughput — across 3 sites, using real-time performance data to drive staffing decisions and reduce operational inefficiencies
- Designed and delivered structured weekly and monthly business intelligence reports for ownership, translating raw site-level data into financial and operational insights that informed executive decisions
- Applied performance trend analysis to manage and coach 20+ employees across multiple locations, identifying execution gaps and improving accountability through data-driven feedback

Site Manager — Mar 2020 – Jan 2022
Xtreme Car Wash · Anniston, AL

- Monitored wash volume, labor utilization, and throughput metrics daily; used demand pattern analysis to optimize scheduling and maintain operational targets

Site Manager — Oct 2017 – Mar 2020
International Car Wash Group · Gadsden, AL

- Managed daily operations and staff performance at a high-volume site, sustaining service quality and operational KPIs through consistent process execution

CERTIFICATIONS

- WGU Specialization: Data Science Professional (Jan 2026)
- WGU Certificate: Data Analytics Professional (Jul 2025)
- WGU Certificate: Data Operations (Sep 2025)
- CompTIA A+ ce

INTERESTS

Large language models & AI · AI agents, orchestration & LLM tooling (Claude, Codex, OpenClaw) · Bitcoin